REV.00

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ENGINE ASS'Y

RECEIVING INSPECTION SHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD 785-7

WO NO:

DISASSEMBLY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

RECEIVING INSPECTION SHEET
ENGINE

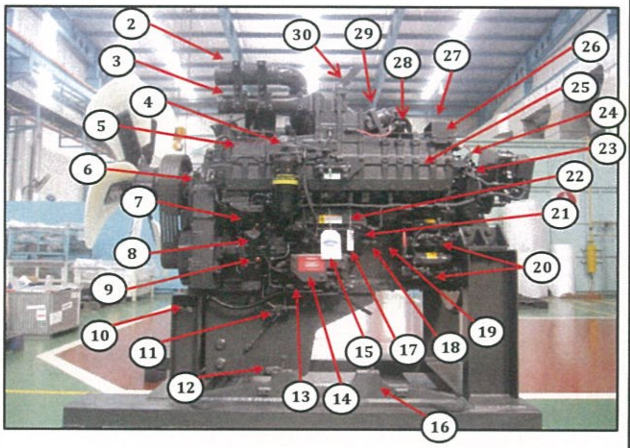
MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	1 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

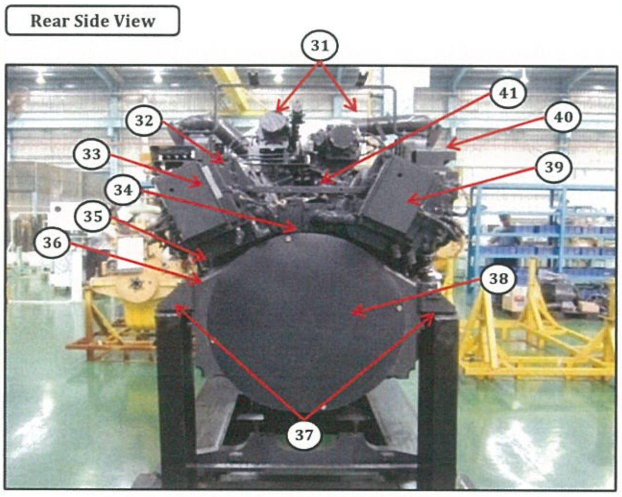
✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES

Left Side View



Rear Side View



NO.	ITEM		INSPECTION CONDITION
1	Painting (Y/B)		
2	Tube Air Intake Connection LH		
3	Air Connector LH		
4	Fuel Filter Assy LH		
5	Engine Name Plate		
6	Tube, Breather With Hose LH		
7	Fuel Supply Pump Assy LH (P/N : 6219-71-1110)		
	S/N :		
8	Tube Oil Filler Cap		
9	Gauge Oil (STD : H+ 0~10 mm)		
	Act : H +....................mm		
10	Engine Stand (Original / Std)		
	Stand No :		
11	Oil Level Sensor Assy		
12	Drain Oil Plug		
13	Priming Pump Assy With Cover LH		
14	Warning Sticker Oil Level		
15	Bracket Mounting		
16	Quality passed Inspection Tag		
17	Caution Tag		
18	Oil Temperature Sensor		
19	Oil Pressure Sensor		
20	Starting Motor Assy (7.5 KW) (P/N : 600-813-7542)		
	Upper S/N :		
	Lower S/N :		
21	Fuel Pressure Sensor With Limiter Assy LH		
22	Common Rail Assy		
23	Boost Pressure Sensor		
24	Intake Temperature Sensor		
25	Air Intake Manifold LH		
26	Exhaust Temperature Sensor (4 pcs) With Bracket		
27	Customer Responsibilities Sticker		
28	Heater Switch Assy With Cover LH		
29	Exhaust Brake Assy LH		
30	Bracket Exhaust Brake Mounting		
31	Turbocharger Assy RH	P/N : 6505-67-5030	
		S/N :	
	Turbocharger Assy LH	P/N : 6505-67-5040	
		S/N :	
32	Heat Tab position		
33	Engine Controller Assy LH		
	S/N :	ESN :	
	P/N Assy :		
34	Caution Flywheel Sticker		
35	Pick Up Revolution Sensor		
36	Flywheel Housing		
37	Rear Bracket Mounting (2 pcs)		
38	Flywheel (P/N : 6219-31-4110)		
39	Engine Controller Assy RH		
	S/N :	ESN :	
	P/N Assy :		
40	Stiker Attention Installation Report		
41	Exhaust Manifold Assy RH & LH		

Notes :

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

RECEIVING INSPECTION SHEET
ENGINE

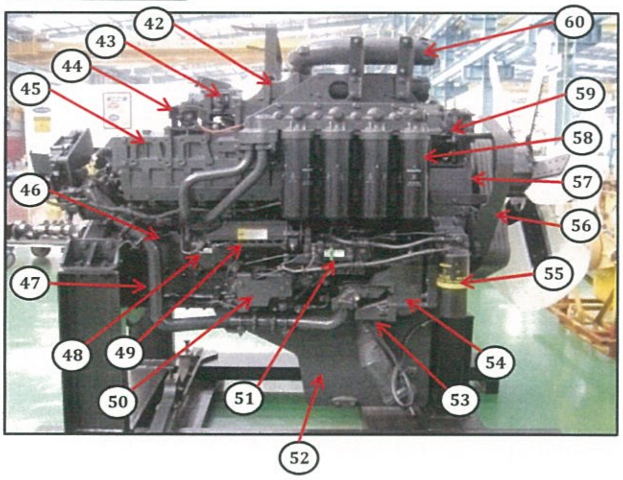
MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	2 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

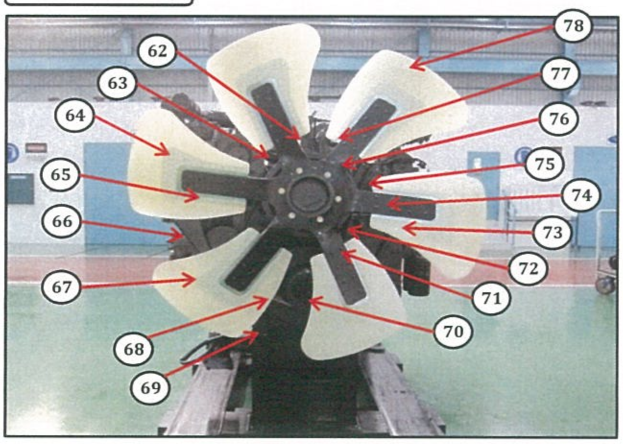
✓ : GOOD CONDITION   X : BAD CONDITION   ⊗ : NONE

SKETCH DRAWING OR PHOTOES

Right Side View



Front Side View



NO.	ITEM	INSPECTION CONDITION
42	Air Connector RH	
43	Exhaust Brake Assy RH	
44	Heater Switch Assy With Cover RH	
45	Air Intake Manifold RH	
46	Oil Cooler Assy	
47	Tube Cooling System Piping	
48	Fuel Pressure Sensor With Limiter Assy RH	
49	Common Rail Assy	
50	Priming Pump Assy With Cover RH	
51	Fuel Supply Pump Assy RH (P/N : 6219-71-1120)	
	S/N :	
52	Oil Pan	
53	Water Inlet Cover Plate	
54	Water Pump Assy	
55	Fuel Filter Assy	
56	Cover Safety Guard Alternator	
57	Tube, Breather With Hose RH	
58	Oil Filter Assy	
59	Alternator Assy (90A) (P/N : 600-825-9330)	
	S/N :	
60	Tube Air Intake Connection RH	
62	Plate Front Hanger	
63	Fan Drive With Pulley Assy	
64	Block Fuel Return With Sensor Assy RH & LH	
65	V-Belt Alternator	
66	Alternator Drive Pulley	
67	Tension Pulley Assy	
68	Vibration Damper Assy (P/N : 6215-31-8200)	
69	Front Support	
70	Crank Pulley Assy	
71	Pointer	
72	Yoke	
73	V-Belt Fan Set	
74	Blowbay Sensor Assy	
75	Water Temperature Sensor	
76	Thermostat Housing	
77	Tube Water Outlet (2 pcs)	
78	Cooling Fan	

Notes :

Inspected by,

Aproved by,

RC/QR04/EG/02/01	REV. 06

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PLANT REBUILD CENTER

ENGINE ASS'Y

SHORT BLOCK ASSEMBLY

CHECK SHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

WO NO. :

SHORTBLOCK

ENGINE ASS'Y SECTION

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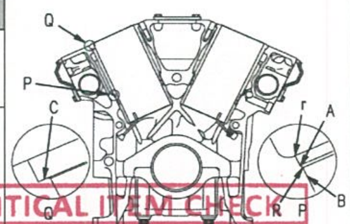
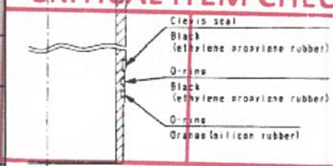
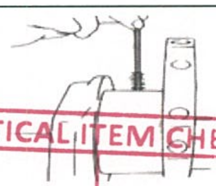
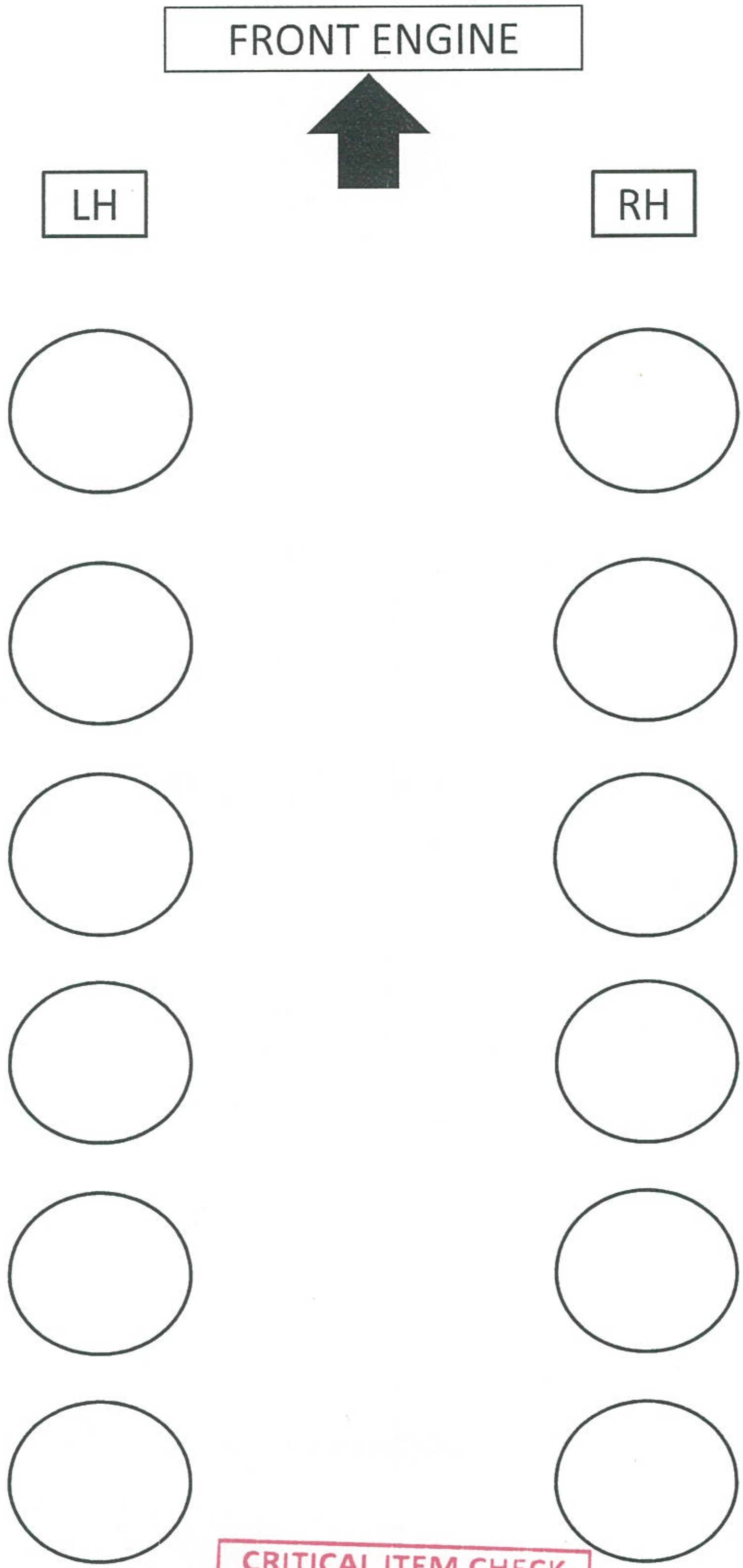
ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	QR03-ENG-011
	ENGINE MODEL		PUBLIC.DATE	28/08/2009
	ENGINE S/N *(new)*		REVISION NO.	06
	MACH.MODEL		PAGE	1 / 6

PROCESS	START/DATE&TIME		ASSEMBLY BY	Short block	
	FINISH/DATE&TIME			Main Line	

Firiing Order		1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L	All dimensions in millimetres
Valve Clereance	Intake	0.35 ± 0.02	All torques in Kilogram metres
	Exhaust	0.57 ± 0.02	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN GL	SIGN SPV	SKETCH DRAWING or PHOTOES
1	Cylinder Block	Bores	Check that the liner counter bore is clean and serviceable prior to fitting liner.					
		Inside of oil hole	Check that all oil holes are free from foreign matter such as chips or dirt.					
		U-plugs & blind plugs	Make sure that all U-plugs and blind plugs have been installed.					
			Make sure that Cyl. block condition is ready to used completely - Tightening plug : Hexagon plug 5 mm | Act : kgm	1.0 ~ 2.0 kgm (Komatsu assembly manual)				CRITICAL ITEM CHECK
			• Oversize head gasket required: Yes | No	THE OTHERS PLUG, SEE PLUG TABLE FOR TIGHTEN				
		S/N	**Cylinder block S/N new:**					Name : | Name :
			Cylinder block S/N original:					QA INSPECTOR | PROD. SUPERVISOR
		Liner deck portion	See that three bondLG-6 has been applied to the proper area.					
2	Cylinder Liner	Crevice seal O-rings	Check that the chamfered side is facing down.					CRITICAL ITEM CHECK 
			Check that the colour of the bottom O-ring is orange					
		Liner	Apply oil immediately before installing. Set the "T" mark on the top surface of the liner facing the front.					Name : | Name :
			Using a cloth, wipe off dirt and oil from contact surface of liner lip and cylinder block counterbore.					QA INSPECTOR | PROD. SUPERVISOR 
		Liner Protrusion	Check the protrusion of the liner: R bank | | | | | | L bank | | | | | |	0.10~0.15mm (Shop Manual Page. 50-5)				
3	Crank shaft	Crank Main Bearings	Make sure that stamp mark on Crank Shaft is available. ***Check Code of Main metal,code are: STD/0.25/0.50*** Code: ____ STD | 0.25 | 0.50 before tightening. Coat the inside face with engine oil.	STD or US 0.25 US 0.50				 CRITICAL ITEM CHECK Name : | Name : QA INSPECTOR | PROD. SUPERVISOR

MEASUREMENT LINER PROTRUSION EACH CYLINDER



CRITICAL ITEM CHECK	
Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

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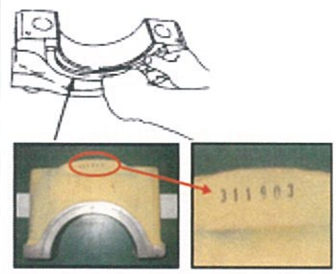
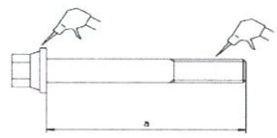
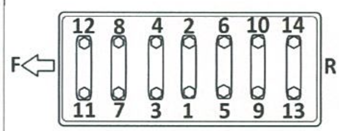
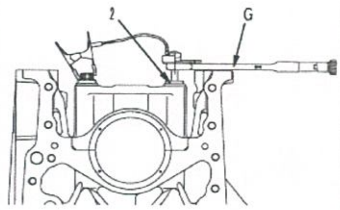
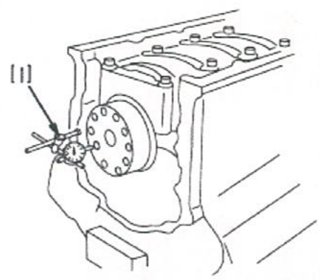
PT. SAPTAINDRA SEJATI

ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3				
	WO NUMBER		**DOCUMENT NO**	QR03-ENG-011
	ENGINE MODEL		**PUBLIC.DATE**	28/08/2009
	ENGINE S/N *(new)*		**REVISION NO.**	06
	MACH.MODEL		**PAGE**	2 / 6

			Do no coat the rear surface of the bearing • Upper bearing has oil hole • Lower bearing has no oil hole • Protrusion of Dowel Pin Act : mm *Make sure the number of main cap same with S/N Engine Block. ***CRANKSHAFT S/N:*** *(Serial No Engine at rear face of Crankshaft)*	**2.7~3.4 mm** (Shop Manual Page. 50-5)				
4	**Crank shaft Cont.**	Thrust bearing	Install the thrust bearing with the groove on the crankshaft side.					
		Torque of Bearing Cap	Bolts must be oiled. When tightening, start from the center and work in order to the Outside Using limit length (**a**) of bolt stem:	**Less than 176.3 mm** (Shop Manual Page. 50-6)				
			CRITICAL ITEM CHECK Name : QA INSPECTOR / Name : PROD. SUPERVISOR Torque sequence mounting bolt main cap.					
			1^{st} step : Act : kgm	**29 ± 1.5 Kgm**				
			2^{nd} step: Act : kgm	**58 ± 1.0 Kgm**				
			3^{rd} step : Act : °	**90 ~120 °** (Shop Manual Page. 50-7)				
			**** After tightening bolt, make 1 punch mark on the head of each bolt. (Don't make a punch mark when using a new bolt)*** **** Punch mark max. 5 punch marks.***					
		Seal Surface	Clean of all dirt and dust on the contact surface of the front and rear seals					
		End Play	Measure crankshaft End Play with dial gauge Act : mm • Tolerance:	**0.140 -0.320 mm** (Shop Manual Page. 50-7)				
			Check that shaft turns smoothly • Standard rotating force: Act : kgm	**1.5 Kgm** (Shop Manual Page. 50-7)				

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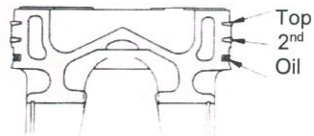
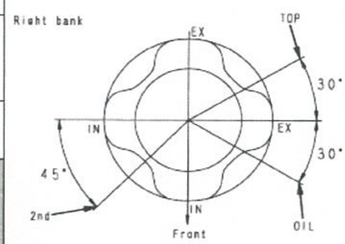
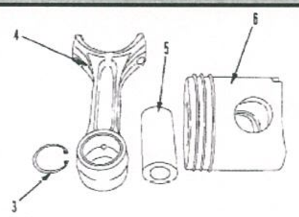
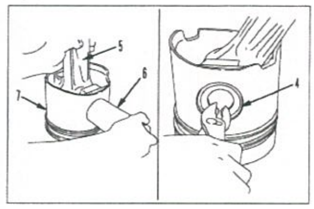
	MACH.MODEL		**PAGE**	3 / 6

	△1	⚠ Radial out & Face Runout Crankshaft	Measure Face & Radial out of Crankshaft at Front and Rear surface Crankshaft before install Con rod assembly.	**Maximum 0.05 mm (std aligment main bearing bore)**				Front Face Runout: mm Front Radial out: mm Rear Face Runout: mm Rear Radial out: mm
5	**Piston and Con. rods**	Piston	When assembling a new connecting rod, mark the cylinder number with an electric punch. Do not make a stamped mark.					
			Set up each ring with the stamped mark facing up. Note the position of the ring gap.	top ring: **1RS** second ring: **2RX**				
			For the oil ring, take out the expander and install it into the piston, then install the oil ring.					
			Confirm that the expander is properly fitted to the groove.					
	⚠	Snap Ring Conrod Bearing	Assemble piston and connecting rod. Confirm that the snap ring which secures the piston pin is properly fitted to the grove. Check that the connecting rod moves smoothly					
		△1	-Check size of Pin metal and recheck Code Pin metal on rear surface of metal.(***Code of Pin Metal are:STD/0.25/0.50***). ☐ Code of Pin metal: | **STD** | **0.25** | **0.50** | | | | |	**STD or US 0.25 US 0.50**				CRITICAL ITEM CHECK Name : | Name : QA INSPECTOR | PROD. SUPERVISOR

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN GL	SIGN SPV	SKETCH DRAWING or PHOTOES
6	**Piston and Con. rods cont**	Con. rod Bearing	Match the number of con. Rod are aligned Check that there is no dirt at the rear face of the bearing before assembling. | CYL | 1 | 2 | 3 | 4 | 5 | 6 | | UP | | | | | | | | LOW | | | | | | | Do not coat the rear surface bearing with oil.					 
	⚠	Inserting Piston & Con. Rod	Check that the hole in the upper bearing and the oil hole in the co. rod are aligned.					
			Set the cylinder block on its side so that the assembly does not enter the cylinder suddenly when inserted.					

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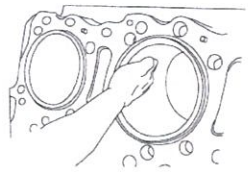
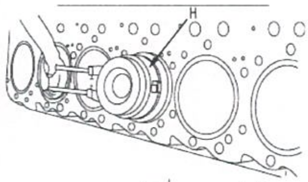
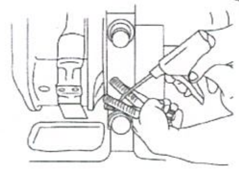
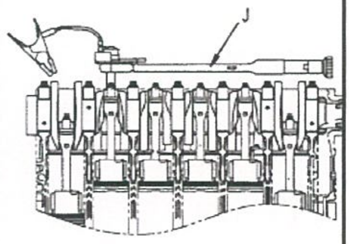
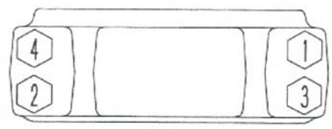
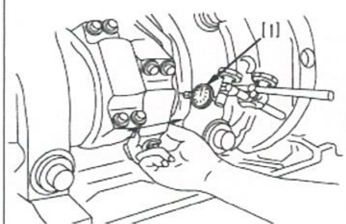
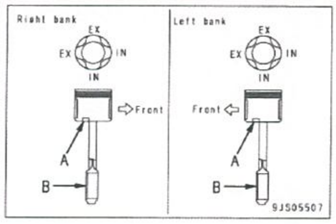
	MACH.MODEL		**PAGE**	4 / 6

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN GL	SIGN SPV	SKETCH DRAWING or PHOTOES
			Coat the cylinder liner with engine oil (SAE 15W-40)					
			uniformly around the whole circumference by Hand.					
			Check that number of piston and connecting rod assembly match cylinder number.					
			Coat connecting rod lower bearing with engine oil, spreading over whole part with finger.					
			Con rod bolts are to have oil (SAE15W-40) on the threads and under the washers.					
		Big End Torque	1st step: Act : kgm	**7 ~ 8 kgm**				
			2nd step: Act : kgm	**7 ~ 8 kgm**				
			3rd step: Act : °	**90^0 ~ 120^0** (Shop Manual Page. 50-11)				
			After tighten of bolt, make 1 punch mark on the head of each bolt (Don't make a punch mark when using a new bolt) * ***Punch mark max. 5 punch marks.***					
			Measure con rod side clearance CYL: 1 / 2 / 3 / 4 / 5 / 6	**0.300 ~ 0.454 mm** (Shop Manual Page. 50-11)				
7	Piston and Con. rods cont.		***Recheck/pastikan bahwa posisi piston sudah benar (dilihat dari posisi top).*** * ***<u>Cowakan yang lebih dalam ada di sisi intake</u>*** * ***<u>Cowakan yang dangkal ada di sisi exhaust</u>***					

CYL	1	2	3	4	5	6

CYL	1	2	3	4	5	6
RH						
LH						

CRITICAL ITEM CHECK

Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

CRITICAL ITEM CHECK

Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

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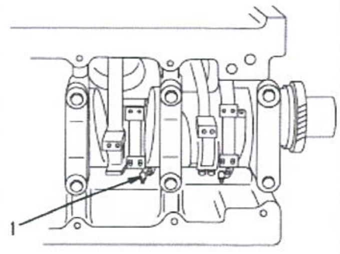
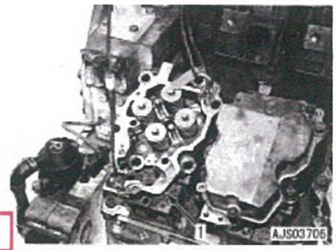
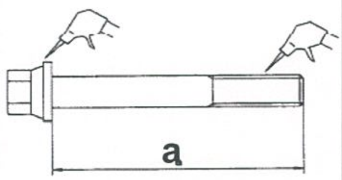
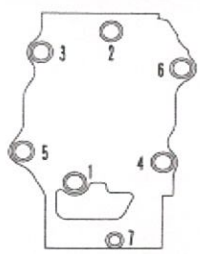
<table>
<tr><td rowspan="4">ASSEMBLY CHECK SHEET
ENGINE ASS'Y
KOMATSU ENGINE 12V140E-3</td><td>WO NUMBER</td><td></td><td>DOCUMENT NO</td><td>QR03-ENG-011</td></tr>
<tr><td>ENGINE MODEL</td><td></td><td>PUBLIC.DATE</td><td>28/08/2009</td></tr>
<tr><td>ENGINE S/N (new)</td><td></td><td>REVISION NO.</td><td>06</td></tr>
<tr><td>MACH.MODEL</td><td></td><td>PAGE</td><td>5 / 6</td></tr>
</table>

<table>
<tr><td></td><td></td><td>Piston Cooling Nozzle</td><td>Check that the piston cooling nozzle is not clogged and that the angle is correct and Install to Block.
Tightening torque
Act : kgm</td><td>5.5 ± 2.0 Kgm
(Shop manual Page. 50-11)</td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="3">8</td><td rowspan="3">Cylinder Head</td><td>Grommets</td><td>- Check that there is no dust or dirt inside the cylinder and on mounting surface.
- When installing the gasket, check that the grommet has been fitted and has not come
- The grommets in the head gasket may become deformed by swelling, so do not coat them with gasket sealant.
- Check and mark grommet with white marker.<table><tr><td>Cyl.</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>RH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>LH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table></td><td rowspan="2">CRITICAL ITEM CHECK
Name :
QA INSPECTOR
Name :
PROD. SUPERVISOR
STD or OVER SIZE</td><td></td><td></td><td></td><td>Cylinder head assembly: 30 kg
</td></tr>
<tr><td>O/S head gasket</td><td>If oversize head gaskets are to be fitted, the block must be stamped O/S.<table><tr><td>STD</td><td>OS</td></tr><tr><td></td><td></td></tr></table></td><td></td><td></td><td></td><td>CODE FOR OVERSIZE GASKET
</td></tr>
<tr><td>Bolts</td><td>-The head bolts must have no dirt in the threads or under the washer, or in the threaded holes in the block.
-Coat the thread and bolt head with Oil. The threaded holes in the block must be free from dirt. A tap can be used to clean the threads</td><td>limit length (a) :
Short bolt =
Less than 170.8 mm
Long bolt =
Less than 205.8 mm</td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="2">No.</td><td rowspan="2">PARTS</td><td rowspan="2">CHECK POINT</td><td rowspan="2">CONDITIONS TO BE CHECKED</td><td rowspan="2">SPEC.</td><td colspan="3">SIGN</td><td rowspan="2">SKETCH DRAWING or PHOTOES</td></tr>
<tr><td>MEC</td><td>GL</td><td>SPV</td></tr>
<tr><td>9</td><td>Cylinder Head Cont.</td><td>Torque</td><td>- Install cyl. head gasket.
- Lift cy. Head ass'y then install to the block
<u>-Recheck bahwa marking nomor di Cyl.Head housing sesuai dgn urutan Pada Cylinder block.</u>
- Tighten Cyl. head mounting bolt .
* No.1- 6 :<table><tr><td>Step 1st</td><td>Act : kgm</td></tr></table></td><td>7 Kgm
14 ~ 16 Kgm</td><td></td><td></td><td></td><td></td></tr>
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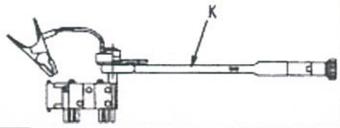
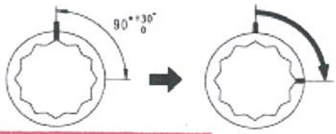
PT. SAPTAINDRA SEJATI

ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	QR03-ENG-011
	ENGINE MODEL		PUBLIC.DATE	28/08/2009
	ENGINE S/N *(new)*		REVISION NO.	06
	MACH.MODEL		PAGE	6 / 6

			Step 2nd Act : kgm Step 3rd Act : kgm Step 4rd Act : kgm	**29 ~ 30 Kgm** **90 ~ 120⁰** (Shop Manual Page. 50-27)				
			*** No.7 :** Act : kgm Torque in the order shown. - After tightening, make a punch mark on the bolt head. If the bolt has 5 marks, do not use it. Replace it with new one.	**6.75 ± 0.75 kgm** (Shop Manual Page. 50-27)				
			- Install the water tube. Water tube joint bolts : Act : kgm	**1.0 ~ 1.3 Kgm** (Shop Manual Page. 50-28)				
			- Install heat label at cylinder head No.6 Act : °	99°c ~ 127°c (210°F ~ 260°F)				

CRITICAL ITEM CHECK

Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

ASSEMBLY BY	INSPECTED & APPROVED BY	
MECHANIC (MEC)	QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION

REV.00

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYL BLOCK

DELIVERY INSPECTION

CHECK SHEET

KOMATSU ENGINE 12V140-3

MACHINE MODEL

WO NO:

SHORTBLOCK SECTION

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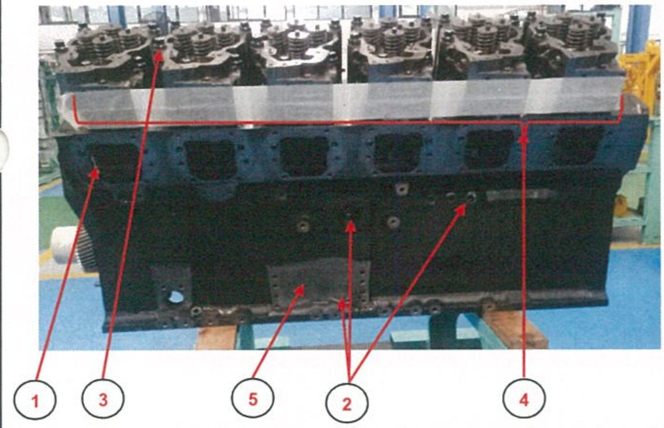
PLANT REBUILD CENTER

DELIVERY INSPECTION SHORT BLOCK ENGINE CHECKSHEET	DOCUMENT NO.	RC/QR06/EG/01/01
	PUBL. DATE	
	REVISION NO.	
	PAGE	1

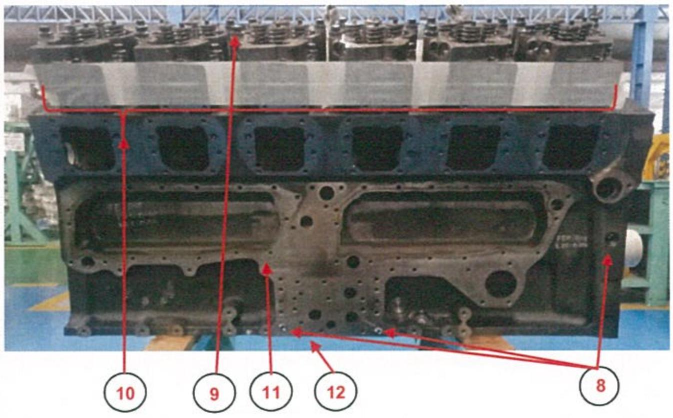
JOB NUMBER		PART NAME	ENGINE ASS'Y
ENGINE MODEL	SAA12V140E-3	DATE / TIME	
ENGINE SER. NO			
MACHINE MODEL	HD 785-7		

SKETCH DRAWING OR PHOTOES

LEFT SIDE



RIGHT SIDE



✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

NO	ITEM (LEFT)	✓	CONDITION	NO	ITEM (RIGHT)	✓	CONDITION
1	Bushing camshaft			7	Bushing camshaft		
	-hole lubricating align looks 60%				-hole lubricating align looks 60%		
	-No scratch and damage				-No scratch and damage		
	-Align bushing camshaft(special tool)				-Align bushing camshaft(special tool)		
	-Conection bushing up side position				-Conection bushing up side position		
2	Hexagon plug			8	Hexagon plug		
3	Bolt cyl head(after +90° turning angle)			9	Bolt cyl head(after +90° turning angle)		
4	Intake hole should be closed(Tape)			10	Intake hole should be closed(Tape)		
5	Surface mounting all component no corotion			11	Surface mounting all component no corotion		
6	All tread bolt id good condition			12	Hexagon and blind plug under cyl block.		

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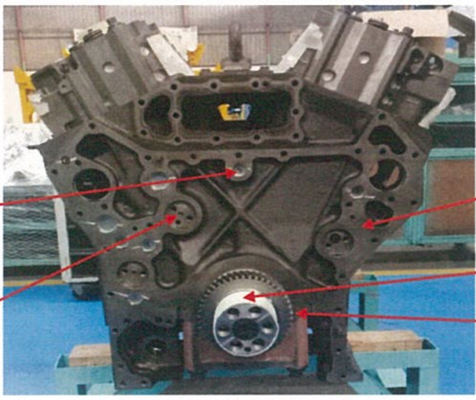
PLANT REBUILD CENTER

DELIVERY INSPECTION SHORT BLOCK ENGINE CHECKSHEET	DOCUMENT NO.	RC/QR06/EG/01/01
	PUBL. DATE	
	REVISION NO.	
	PAGE	2

JOB NUMBER		PART NAME	ENGINE ASS'Y
ENGINE MODEL	SAA12V140E-3	DATE / TIME	
ENGINE SER. NO			
MACHINE MODEL	HD 785-7		

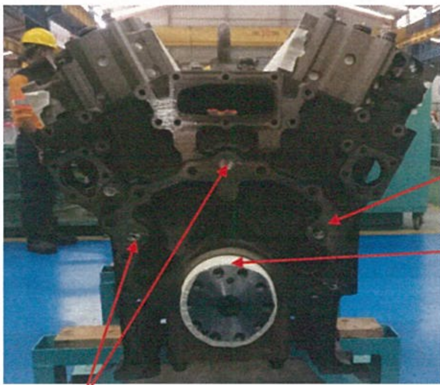
SKETCH DRAWING OR PHOTOES

FRONT



13 14 15 16 17

REAR



18 19 20

✓: GOOD CONDITION X : BAD CONDITION ⊗ : NONE

FRONT				REAR			
NO	ITEM	✓	CONDITION	NO	ITEM	✓	CONDITION
13	Timing gear hole(no damage and corrotion)			18	Blind & Hexagon plug		
14	Blind & Hexagon plug			19	Surface for mounting all component no damage&corotion		
15	Surface for mounting all component no damage&corotion						
16	Frontseal surface crankshaft (should be closed tape)			20	Rear seal surface crankshaft (should be closed tape)		
17	Gear crankshaft not scratch and pitcing						

DELIVERY BY SHORT BLOCK SECTION
DELIVERED BY :
MACHINIST NAME :
SIGN

RECEIVING INSPECTION BY MAINLINE SECTION
RECEIVED DATE :
MECHANIC NAME :
SIGN

REV.01

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ENGINE ASS'Y

DELIVERY AND INSPECTION CHECK SHEET

KOMATSU ENGINE 12V140-3

MACHINE MODEL

HD 785-7

JOB NO:

ENGINE ASS'Y SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

DELIVERY INSPECTION SHEET
ENGINE

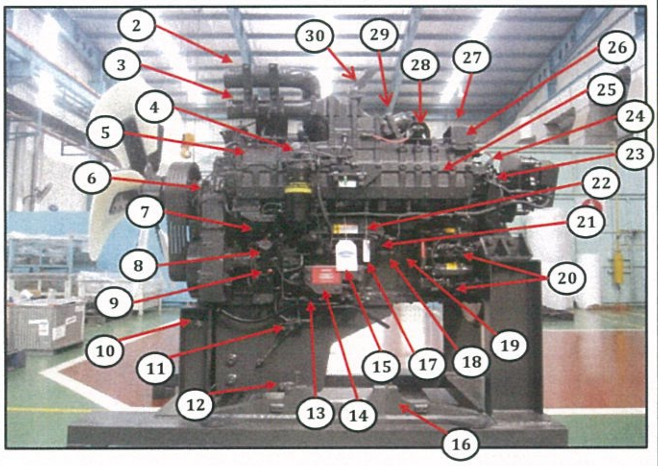
MACHINE MODEL	HD785-7	ENGINE PART NO.	6219-B0-0031	DOCUMENT NO.	RC/QR06/EG/24/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	7-Sep-10
ENGINE NO.		DELIVERY DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	1 OF 3

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

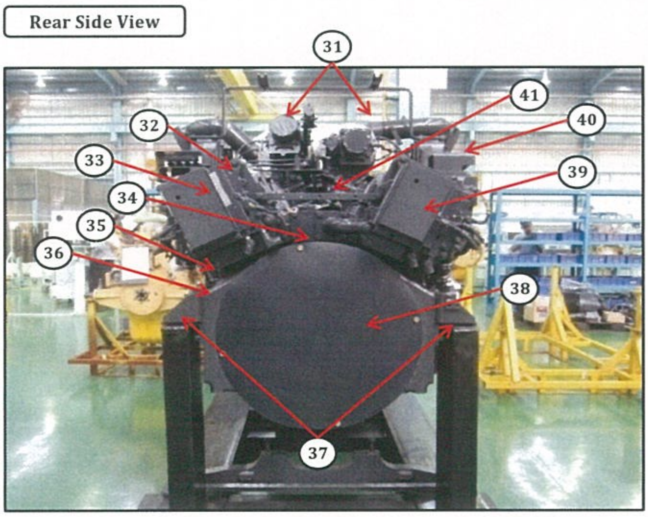
✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES

Left Side View



Rear Side View



NO.	ITEM	INSPECTION CONDITION
1	Painting (Y/B)	
2	Tube Air Intake Connection LH	
3	Air Connector LH	
4	Fuel Filter Assy LH	
5	Engine Name Plate	
6	Tube, Breather With Hose LH	
7	Fuel Supply Pump Assy LH (P/N : 6219-71-1110)	
	S/N :	
8	Tube Oil Filler Cap	
9	Gauge Oil (STD : H+ 0~10 mm)	
	Act : H +....................mm	
10	Engine Stand (Original / Std)	
	Stand No :	
11	Oil Level Sensor Assy	
12	Drain Oil Plug	
13	Priming Pump Assy With Cover LH	
14	Warning Sticker Oil Level	
15	Bracket Mounting	
16	Quality passed Inspection Tag	
17	Caution Tag	
18	Oil Temperature Sensor	
19	Oil Pressure Sensor	
20	Starting Motor Assy (7.5 KW) (P/N : 600-813-7542)	
	Upper S/N :	
	Lower S/N :	
21	Fuel Pressure Sensor With Limiter Assy LH	
22	Common Rail Assy	
23	Boost Pressure Sensor	
24	Intake Temperature Sensor	
25	Air Intake Manifold LH	
26	Exhaust Temperature Sensor (4 pcs) With Bracket	
27	Customer Responsibilities Sticker	
28	Heater Switch Assy With Cover LH	
29	Exhaust Brake Assy LH	
30	Bracket Exhaust Brake Mounting	
31	Turbocharger Assy RH P/N : 6505-67-5030	
	S/N :	
	Turbocharger Assy LH P/N : 6505-67-5040	
	S/N :	
32	Heat Tab position	
33	Engine Controller Assy LH	
	S/N : ESN :	
	P/N Assy :	
34	Caution Flywheel Sticker	
35	Pick Up Revolution Sensor	
36	Flywheel Housing	
37	Rear Bracket Mounting (2 pcs)	
38	Flywheel (P/N : 6219-31-4110)	
39	Engine Controller Assy RH	
	S/N : ESN :	
	P/N Assy :	
40	Stiker Attention Installation Report	
41	Exhaust Manifold Assy RH & LH	

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

DELIVERY INSPECTION SHEET
ENGINE

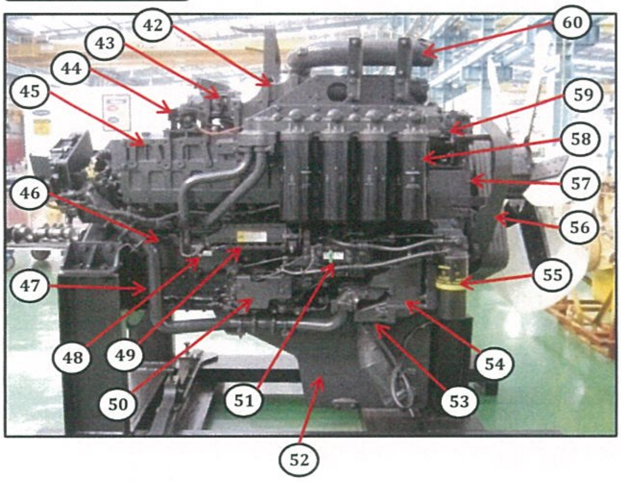
MACHINE MODEL	HD785-7	ENGINE PART NO.	6219-B0-0031	DOCUMENT NO.	RC/QR06/EG/24/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	7-Sep-10
ENGINE NO.		DELIVERY DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	2 OF 3

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

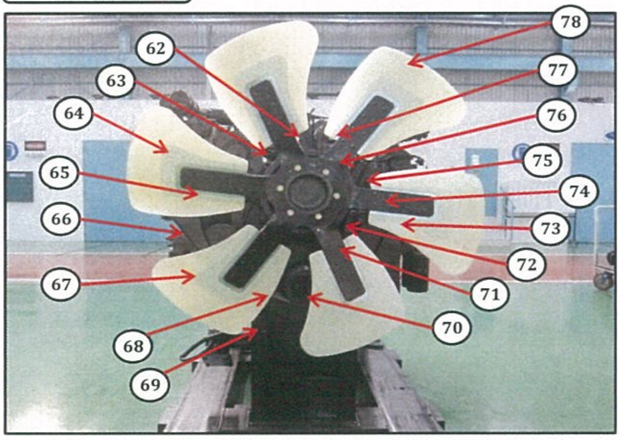
✓ : GOOD CONDITION   X : BAD CONDITION   ⊗ : NONE

SKETCH DRAWING OR PHOTOES

Right Side View



Front Side View



NO.	ITEM	INSPECTION CONDITION
42	Air Connector RH	
43	Exhaust Brake Assy RH	
44	Heater Switch Assy With Cover RH	
45	Air Intake Manifold RH	
46	Oil Cooler Assy	
47	Tube Cooling System Piping	
48	Fuel Pressure Sensor With Limiter Assy RH	
49	Common Rail Assy	
50	Priming Pump Assy With Cover RH	
51	Fuel Supply Pump Assy RH (P/N : 6219-71-1120)	
	S/N :	
52	Oil Pan	
53	Water Inlet Cover Plate	
54	Water Pump Assy	
55	Fuel Filter Assy	
56	Cover Safety Guard Alternator	
57	Tube, Breather With Hose RH	
58	Oil Filter Assy	
59	Alternator Assy (90A) (P/N : 600-825-9330)	
	S/N :	
60	Tube Air Intake Connection RH	
62	Plate Front Hanger	
63	Fan Drive With Pulley Assy	
64	Block Fuel Return With Sensor Assy RH & LH	
65	V-Belt Alternator	
66	Alternator Drive Pulley	
67	Tension Pulley Assy	
68	Vibration Damper Assy (P/N : 6215-31-8200)	
69	Front Support	
70	Crank Pulley Assy	
71	Pointer	
72	Yoke	
73	V-Belt Fan Set	
74	Blowbay Sensor Assy	
75	Water Temperature Sensor	
76	Thermostat Housing	
77	Tube Water Outlet (2 pcs)	
78	Cooling Fan	
79	RFID Tag, Traceability mounting bolt	
80	Masking condition	
81	Check Mounting Bolt (Rear Bracket & Front Support)	
	Std Tightening Torque : 25 kgm	

Notes :

Inspected by,

Aproved by,

REV.06

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SERVICES
PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ENGINE ASS'Y

MAINLINE ASSEMBLY CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

WO NO:

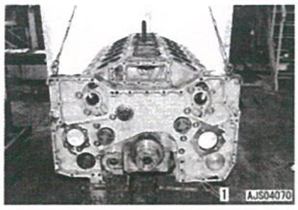
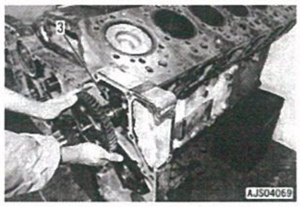
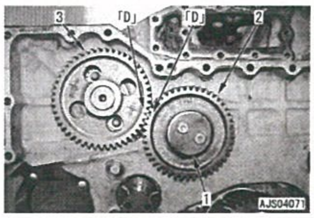
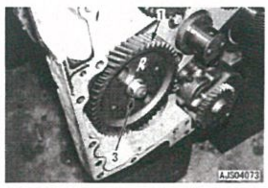
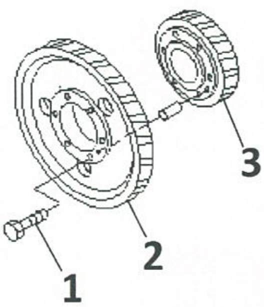
ENGINE ASS'Y SECTION

adaro SERVICES
PT. SAPTAINDRA SEJATI

MAIN ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3				
	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC.DATE	28/08/2009
	ENGINE S/N *(new)*		REVISION NO.	06
	MACH.MODEL	HD 785-7	PAGE	1 / 12

PROCESS	START/DATE&TIME		ASSEMBLY BY	Short block	
	FINISH/DATE&TIME			Main Line	

Firiing Order		1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L	All dimensions in millimetres
Valve Clereance	Intake	0.35 ± 0.02	All torques in Kilogram metres
	Exhaust	0.57 ± 0.02	

No.	PARTS	CHECK	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN GL	SIGN SPV	SKETCH DRAWING or PHOTOES
1	Timing Gear Case		- Prior to installation, fix several portions on gasket temporarily using the gasket sealant so that it may be displaced. Mounting Bolt: Act : kgm	10.0 ~ 12.5 kgm (Shop Manual page. 12-06)				
2	Camshaft		- Turn the cam gear and tighten the mounting bolt, through lightening hole. * **Replace the current mounting bolts with a new one** - Cam shaft mounting bolt : Act : kgm	9.0 ± 3.5 Kgm (Shop Manual page. 50-12)				
3	Idler gears	Sub Idler gear Torque Bolt	- When install idler gear, align the counter mark" D" to that the cam gear for the right bank. - Coat the idler gear bushing inner surface with engine oil (15W40). - Install plate (1) and tighten mounting bolts. - Tightening torque : Act : kgm	10.0 ~ 12.5 kgm (Shop Manual Page. 50-14)				
		Supply pump gear Torque Nut	- Install key (2) to the shaft - Install supply pump gear and tighten mounting nut. **Coat on thread of shaft supply pump use adesive Locktite LT2(High strength type)* - Tightening torque mounting nut : Act : kgm	18 ~ 20 kgm (Shop Manual Page. 50-14)				
		Idle gear (large)	- Install pin dowel at Idle gear (small)(3), - Install idle gear (large)(2) to idle gear(small),match pin dowel idle gear(small)(3) to hole idle gear(large)(2). Mounting bolt(1) tighten: Act : kgm	6.0~7.5 Kgm				

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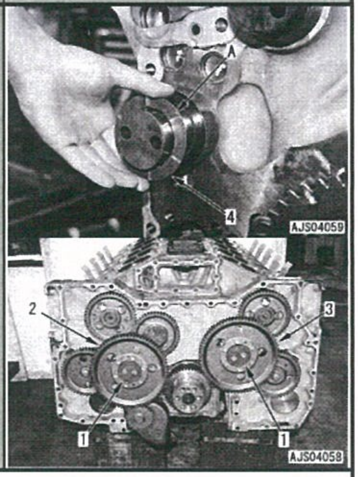
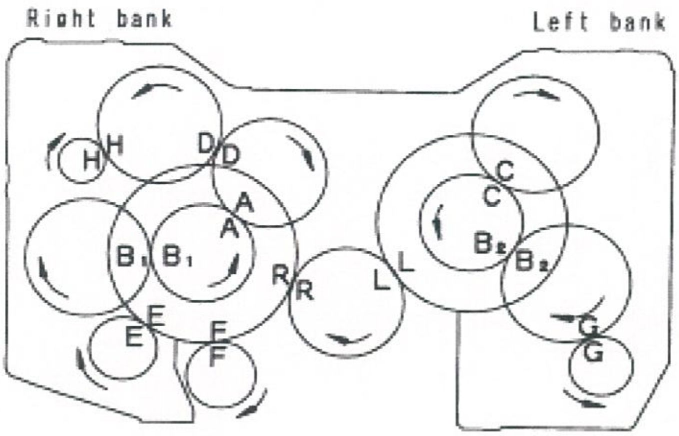
SERVICES
PT. SAPTAINDRA SEJATI

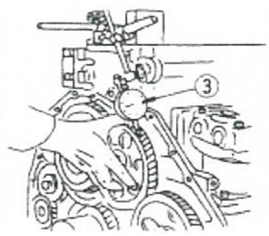
MAIN ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC.DATE	28/08/2009
	ENGINE S/N *(new)*		REVISION NO.	06
	MACH.MODEL	HD 785-7	PAGE	2 / 12

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN GL	SIGN SPV	SKETCH DRAWING or PHOTOES
4	Idler Gears Cont.	Idler gear (large)	- Set the No.1 cylinder head on the right bank to the top dead center position. - Coat the idler gear bushing inner surface with engine oil (15W40). - Install trust plate (4) to the shaft. - Install idler gear (2) and (3), then tighten mounting bolts. - Tightening torque mounting bolts : Act : kgm ***Install gear such the counter marks as well as that of respective gear may come to the positions shown on the figure.**	**10.0 ~ 12.5 kgm** (Shop Manual Page. 50-15)				



CRITICAL ITEM CHECK

Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

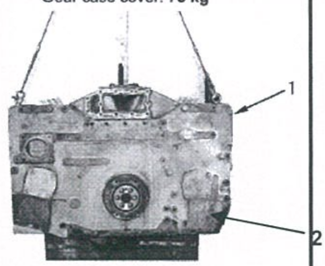
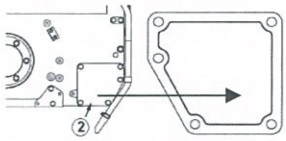
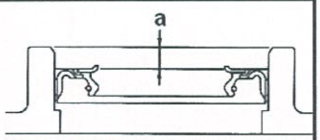
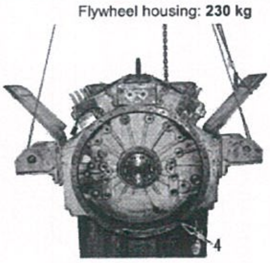
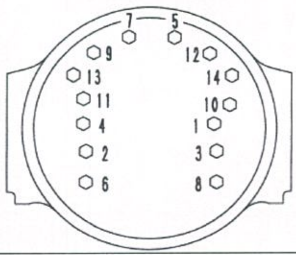
Timing Gear Marked	Check Point	Backlash (mm) Standard	Backlash (mm) Measured		SIGN MEC	SIGN GL	SIGN SPV	SKETCH DRAWING or PHOTOES
A • A	Right idler gear (small) and right sub idler gear.	0.130 ~ 0.390 (0.125 ~ 0.363)						**Note :** **The standard value enclosed in ( ) indicates the backlash between " spur gears"**
B1 • B1	Right sub idler and right supply pump gear.	0.051 ~ 0.469 (0.125 ~ 0.363)						
B2 • B2	Left sub idler and right supply pump gear.	0.051 ~ 0.469 (0.125 ~ 0.363)						
C • C	Left sub idler gear and left cam gear	0.129 ~ 0.391 (0.125 ~ 0.363)						
D • D	Right idler gear (small) and right cam gear.	0.129 ~ 0.391 (0.125 ~ 0.363)						
E • E	Right idler gear (large) and water pump drive gear.	0.052 ~ 0.481						
F • F	Right idler gear (large) and oil pump gear.	0.137 ~ 0.421						
G • G	Left supply pump gear and compressor gear	0.106 ~ 0.381 (0.138 ~ 0.354)						
H • H	Right cam gear and accessory drive gear	0.085 ~ 0.492 (0.138 ~ 0.354)						
L • L	Crank gear and left idler gear (large)	0.141 ~ 0.425						
R • R	Crank gear and right idler gear (LARGE)	0.141 ~ 0.425						
Check Point		End Play (mm)			SIGN			

CRITICAL ITEM CHECK

Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

MAIN ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC.DATE	28/08/2009
	ENGINE S/N *(new)*		REVISION NO.	06
	MACH.MODEL	HD 785-7	PAGE	3 / 12

Check Point		End Play (mm)		SIGN		
		Standard	Measured	MEC	GL	SPV
1	Idler gear (large)(LH) end play	0.07~ 0.18				
2	Idler gear (large)(RH) end play	0.07~ 0.18				
3	Idler gear (small) end play	0.07~ 0.18				
4	Supply pump gear(LH) end play	0.07~ 0.20				
5	Supply pump gear(RH) end play	0.07~ 0.20				
6	Oil pump gear end play	0.03 ~0.088				
7	Accessory drive gear end play	0.1 ~0.4				
8	Crank gear end play	0.14 ~ 0.032				
9	Cam gear(LH) end play	0.10 ~ 0.25				
10	Cam gear(RH) end play	0.10 ~ 0.25				

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN GL	SIGN SPV	SKETCH DRAWING or PHOTOES
5	Gear Case Cover		- Fit gasket, install gear case cover. Torque mounting bolt: Act: kgm	**10 ~ 12.5 kgm**				Gear case cover: **75 kg** 
			*** When cover (2) was removed, apply the gasket sealant in the position shown in the figure before install again.** ****Use Three bond 1207D spec.*** 					
		Front Seal	- Check front seal specification S/W | STD | OS - Press in the front seal uniformly so that it is not at an angle.	**STD or OVER SIZE** S = Single Lip Seal W= Double Lip Seal				
6	Flywheel Housing	assembly	- No crack and no damage - No damage thread hole - Coat with gasket sealant(**Three bond 1215 spec**) then raise housing and install - Coat mounting bolt with engine oil SAE15W~40 - Tighten the flywheel housing mounting bolts in order from diagram. Oil bolt threads.					Flywheel housing: **230 kg** 
			1st step: Act : kgm	**35± 5 kgm**				
			2nd step:	**Loosen it**				
			3rd step: Act : kgm	**42± 2 kgm** (Shop Manual Page. 50-20)				

Front seal check:

S/W	STD	OS

Dimension b	Bolt hole 6-12 x 1.75	Oil seal press-fitting depth **a**
109 mm	Absent	13 – 14 mm
114 mm	Present	18 – 19 mm

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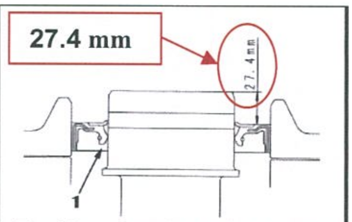
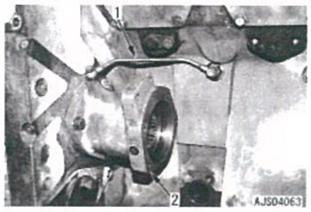
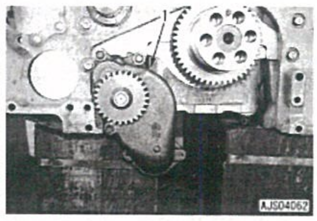
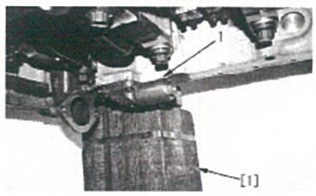
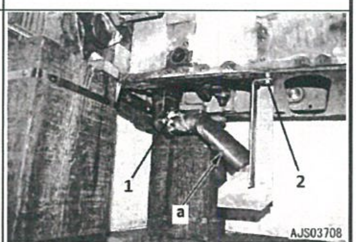
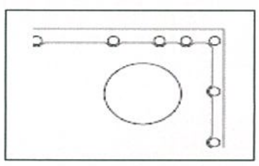
7		Rear Seal	S/W | STD | OS - Make sure that the rear seal no pitting, no scratch. - Press fit the rear seal (1)by using installation tool to obtain correct position. Be careful not to damage the seal when installing the housing.	**STD or OVER SIZE** S= Single Lip W= Double Lips				 For New Crank shaft = **20mm**
No.	**PARTS**	**CHECK POINT**	**CONDITIONS TO BE CHECKED**	**SPEC.**	**SIGN** EC	GL	SPV	**SKETCH DRAWING or PHOTOES**
8	**Supply pump Drive Case**		- Install supply pump drive case Tigthen mounting bolt: LH Bank Act kgm	**6.0~7.5 Kgm**				
			RH Bank Act kgm	**10 ~ 12.5 kgm**				
			- Install oil tube. *** Install oil tube on the left bank alone.** - Oil tube joint bolts : Act : kgm	**0.8 ~ 1.5 Kgm** (Shop Manual Page. 50-13)				
9	**Oil Pump Assy**	**Pump**	Oil Pump: New | Reuse - Check all of pump condition - Fill in oil engine to inner of pump - Rotate pump by hand smoothly - Fit oring for sensor side					
		1	- Mounting oil pump and tighten mounting bolts. Act : kgm	**10 ~ 12.5 kgm** (Shop Manual Page. 00-32)				
		Main Relief Valve	- Recheck plug of main relief valve (retighten) Relief Valve: New | Reuse					
		Strainer Suction pipe	Install strainer to oil pump(a),Fit oring and tighten mounting bolt. 1 Act: kgm 2 Act: kgm	**10 ~ 12.5 kgm** **6.0~7.5 Kgm** (Shop Manual Page. 00-32)				
10	**Oil Pan**		Oil Pan: New | Reuse - Recheck Condition 0f Oil Pan (No crack , No band and No damage) - Install the gasket and push up oil pan					

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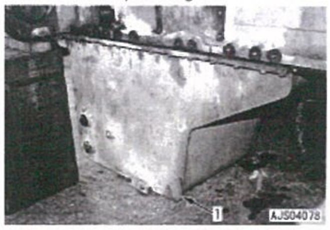
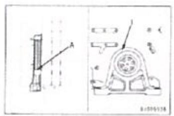
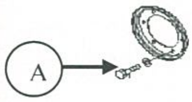
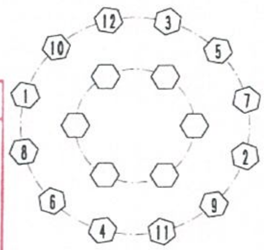
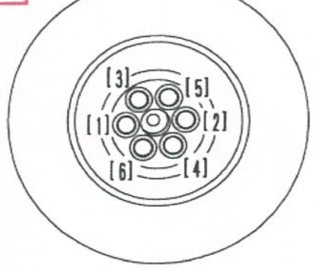
	Oil Pan		with a pallet lift, etc. and tighten the mounting bolts - Connect the level sensor connector. - When tightening, start from center and work in order to the out side. - Tighten mounting bolts. Act : kgm	**6.0~7.5 Kgm** (Shop Manual Page. 00-32)				Oil pan: **75 kg** 

No.	**PARTS**	**CHECK**	**CONDITIONS TO BE CHECKED**	**SPEC.**	**SIGN**			**SKETCH DRAWING or PHOTOES**
					MEC	**GL**	**SPV**	
11	**Front Suport**		- Prior to the installation, fill the oil hole with Grease. - Volume of grease filled : 20 cc - Tighten mounting bolts cover (A) Act : kgm	**STD Grease :** **(G2-LI)** **10 ~ 12.5 Kgm** (Shop Manual Page. 00-32)				 
12	**Dumper and Crank Pulley Ass'y**	Damper	- Lift and install damper and crank pulley assembly, * **Install the assembly aligning it to the dowel pin of the crankshaft.** - Coat mounting bolts and washer with engine oil **SAE15W40** - Tightening Mounting bolt: Act : kgm	**10 ~ 12.5 kgm** (Shop Manual page. 50-19)				
13	**Dumper and Crank Pulley Ass'y Cont.**	Damper	- Tighten the mounting bolts in 3 steps according to the order indicated in the figure. - Coat mounting bolts and washer with engine oil **SAE15W40** - Tightening Mounting bolt: Step 1st — Act : kgm Step 2nd — Act : kgm Step 3rd — Act : kgm	**7.5 ± 2 Kgm** **25 ± 2 Kgm** **76 ± 2 Kgm** (Shop Manual page. 50-19)				

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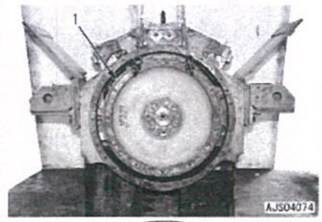
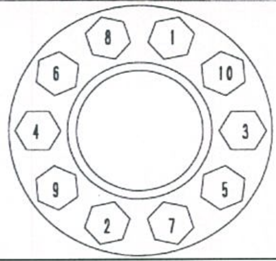
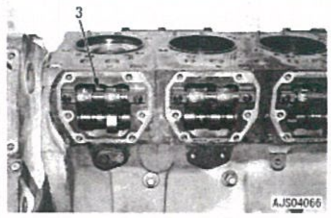
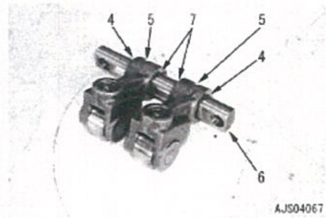
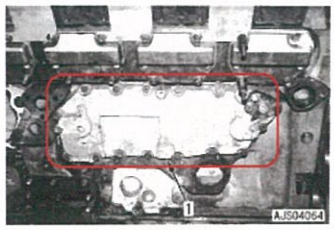
14	Flywheel		-Using eye bolt @ (Dia. = 12 mm, Pitch = 1.75 mm), -Sling flywheel , install the flywheel to the crankshaft, -Coat the threads of the bolts and the seat face, with engine oil 15W40 * **Make a punch mark on the bolt head each time the bolts are used.** Fly wheel mounting bolts : Step 1st | Act : kgm Step 2nd | Act : kgm Step 3rd | Act : kgm	**10 ± 2 Kgm** **30 ± 2 Kgm** **55 ± 2 Kgm** (Shop Manual page. 50-23)				Flywheel assembly: **130 kg**  
15	Cam Follower	assembly	- Before installing the cam follower ass'y, Wash the oil hole of the mounting bolt. - Make sure that the oil hole is free from dirt. - Use LG-6 liquid gasket on the cover face. (Use Three bond 1215 spec)	**4.5 ~ 6.0 kgm** (Shop Manual page. 50-13)				
	Cam Follower	Torque	- Cam follower mounting bolt . Act : kgm - Cam follower cover mounting bolts. Act : kgm	**3.5 ~ 4.5 Kgm** (Shop Manual Page. 50-13)				
No.	**PARTS**	**CHECK POINT**	**CONDITIONS TO BE CHECKED**	**SPEC.**	**SIGN** **MEC**	**QA**	**SPV**	**SKETCH DRAWING or PHOTOES**
16	Oil Cooler Ass'y		Oil cooler | New | Reuse Front Side | | Rear Side | | - Check that there is no dust or dirt inside the oil cooler and no crack - Fit gasket and install oil cooler assembly - Oil Cooler mounting bolt: Act : kgm -Use gasket sealant Three bond 1215 spec.	**(M10) Bolts : 6.0 ~ 7.5 Kgm** (Shop Manual page. 00-32)				

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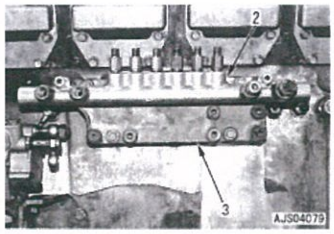
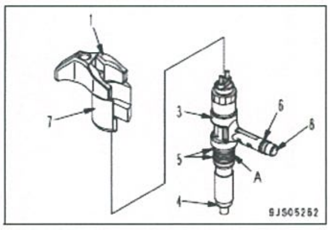
No	Item	Part	Procedure	Standard				Picture
17	**Common Rail**		- Install Bracket (3) - Install common rail (2) -Common rail mounting bolts : Act : kgm *** Install the common rail at this point since positioning af the high-pressure pipe needed when installing the fuel injector.**	**4 ~ 5 kgm** (Shop Manual Page. 50-26)				
18	**Heat tab**		- Install the water tube. Water tube joint bolts : Act : kgm	**1.0 ~ 1.3 Kgm** (Shop Manual Page. 50-28)				
			- Install heat label at cylinder head No.6 Act : °	99°c ~ 127°c (210°F ~ 260°F)				
21	**Valve mechanis**	Cross Heads	- Fit cross heads and adjust so that when the screw has touched the valve stem it is advanced 20°. - Cross head adjusting screws go to exhaust manifold side of head. - Lock nut tighten torque: Act : kgm	**6.0 ± 0.6Kgm** (Shop Manual Page. 50-28)				
		Push rod	- Install Push rod in the hole * Check that the push rod is in the cam follower.					
	Valve mechanis	Fuel injector	- Check that the inside of the injector Sleeve is free from dirt. ***O-ring : coat with Engine oil** ***Ball washer: coat with Engine oil** -Holder mounting bolt: Act = kgm Write injectors code below:	**6.0 ~ 7.5 kgm** (Shop Manual Page. 50-28)				

CYL.NO	INJECTOR CODE	HOLE OF NOZZLE
1R		
2R		
3R		
4R		
5R		
6R		
1L		
2L		
3L		
4L		
5L		
6L		

**Example*: Injector code 6290 05M 0128
Hole of nozzle: 8 Holes

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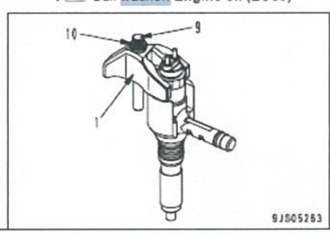
3) Fit ball washer (10) to bolt (9) and then tighten injector holder (1) temporarily.
Ball washer: Engine oil (EO30)



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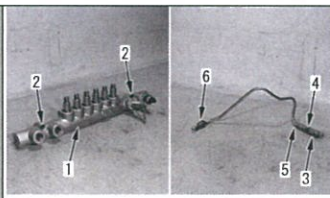
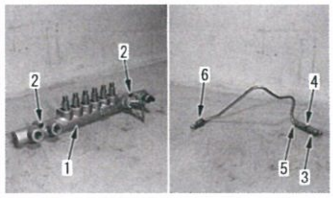
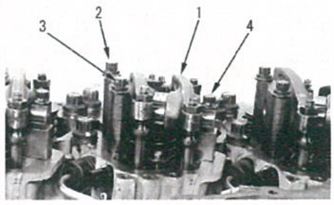
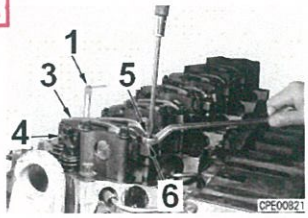
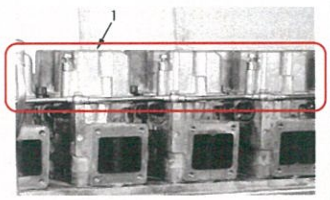
		Common rail and high-pressure pipe	- Before installing the high-pressure pipe, check it for the following defects. If there is any of these defects, it can cause fuel leakage. Accordingly, replace the high-pressure pipe.					
22	**Valve Mechanis**	Torque	- Tighten special bolts (2) : Act : kgm - Tighten Sleeve Nut (5)(6) : Act : kgm - Tighten Clamp : Act : kgm	**6.02 ~ 7.55kgm** (Shop Manual Page. 50-16) **4 ~ 5 kgm** (Shop Manual Page. 50-17) **1.0 ± 0.1 kgm** (Shop Manual Page. 50-17)				
		Rocker Arm and Shaft Ass'y	-Install rocker arm and shaft assembly (1) and -tighten mounting bolts (2), (3) and (4). *Before tightening the mounting bolts, check that the ball of the adjustment screw is fitted to the socket of the push rod. - TightenTorque mounting bolts. Act : kgm	**9.5 ~10.5 kgm** (Shop Manual Page. 50-17)				
	Valve Mechanis	AdjustingValve clereance	-Adjust valve clereance Intake : Act : mm	**0.35 ± 0.02** (Shop Manual Page. 13-39)				
			-Adjust valve clereance. Exhaust : Act : mm	**0.57 ± 0.02** (Shop Manual Page. 50-17)				
			-Tighten locknut (6) to secure adjustment screw (5) -Locknut: Act : kgm	**5.4 ~6.6 kgm** (Shop Manual Page. 50-18)				
		Head cover	-Fit the O-ring and install cylinder head cover. -Since the bolt at the front on the air intake manifold side is also used to secure the wiring harness clamp, tighten it temporarily. -Mounting bolt: Act : kgm	**3.0 ~ 3.5 kgm** (Shop Manual Page. 50-18)				

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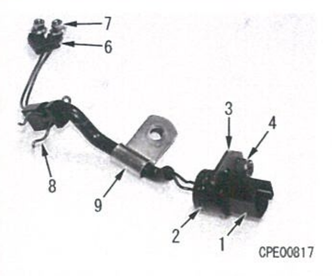
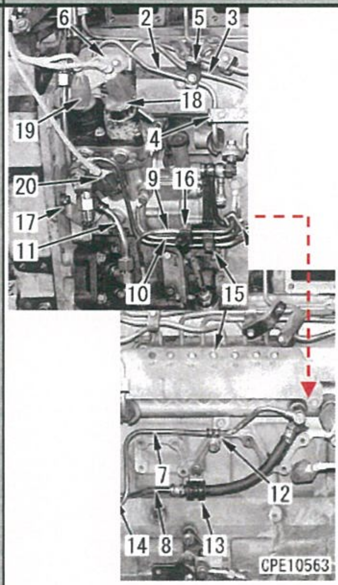
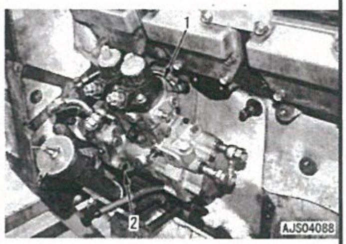
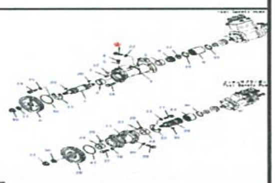
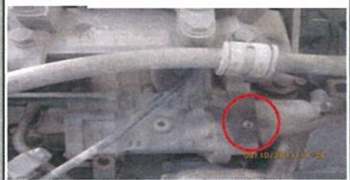
23	**Wiring Harness & Sensor Connector**		- Install wiring harness terminal (6) to fuel injector (5) and tighten 2 nuts (7 - Terminal nut tighten torque: Act : kgm - Install sensor connector	**0.2 ± 0.02 kgm** (Shop Manual page. 50-22)				
No.	**PARTS**	**CHECK POINT**	**CONDITIONS TO BE CHECKED**	**SPEC.**	**SIGN** **MEC**	**QA**	**SPV**	**SKETCH DRAWING or PHOTOES**
24	**Piping & wiring of supply pump**		- Install bracket (1). - install fuel tubes (2) and (3). Sleeve nut of tube : • *Pump side* Act : kgm • *Common rail side* Act : kgm Clamping bolt : Act : kgm - Install fuel tubes (7) and (8). • *Pump side* Act : kgm •*Common rail side* Act : kgm - Install Oil tubes (9) (10) and (11). • *Joint bolt of tubes (9) and (10).* Act : kgm • *Sleeve nut of tube (11)* Act : kgm - Mounting bolt of clamps (14), (15) and (16): Act : kgm **- Mounting supplypump(1):** Act : kgm **-Supply pump assy S/N:** **RH:** **LH:**	**4 ~ 5 kgm** (Shop Manual page. 50-23) **4 ~ 6 kgm** (Shop Manual page. 50-23) **1.0 ± 0.1 kgm** (Shop Manual page. 50-23) **1.5 ~ 2.0 kgm** (Shop Manual page. 50-23) **1.8 ~ 2.3 kgm** (Shop Manual page. 50-23) **0.8 ~ 1.3 kgm** (Shop Manual page. 50-23) **4.4 ~4.8 kgm** (Shop Manual page. 50-23) **0.8 ~ 1.3 kgm** (Shop Manual page. 50-23) **6.7 ± 0.7 kgm** (Shop Manual page. 50-23)				
25	**Lock bolt**	△1	-Mounting Lock Bolt on Fuel pump drive RH and LH and install the washer gasket. Act: kgm *(after finish install fuel pump assy to fuel Pump drive).*	**0.8 ~ 1.2 kgm** (Shop Manual page. 50-23)				

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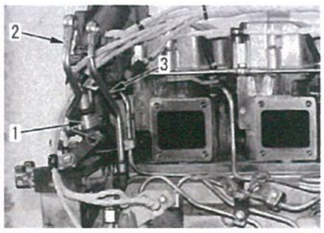
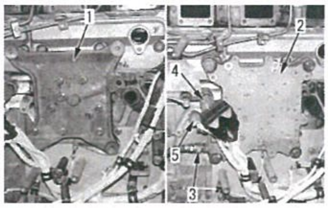
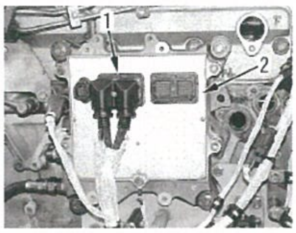
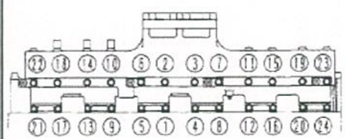
No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	SKETCH DRAWING or PHOTOES
26	**Breather remote tube and corrosion resistor tube**		- Install breather remote tube (3). - Install corrosion resistor tubes (2) and (1) temporarily					
27	**Engine controller cooler**		- Install the O-ring and plate (1). - Install engine controller cooler (2) and fuel hose (3) ****Install the hose on the engine controller cooler inlet side when installing the fuel filter head.*** - Cooling plate mounting bolt: Act : kgm - Connect fuel pressure sensor connector PFUEL (4) and engine oil pressure sensor connector POIL (5)	**3.5 ~ 4 kgm** (Shop Manual page. 50-32)				
No.	**PARTS**	**CHECK POINT**	**CONDITIONS TO BE CHECKED**	**SPEC.**	**SIGN MEC**	**SIGN QA**	**SIGN SPV**	**SKETCH DRAWING or PHOTOES**
28	**Engine controller**		- Install engine controller (2) Torque mounting bolt: Act : kgm And conect conector main harnest(1) **-Engine controller S/N:** **RH:** **LH:**	**2,4 kgm** (Shop Manual page. 50-38)				
29	**Air Intake Manifold**		- Fit the gasket and sling and install air intake manifold (1) and then tighten the mounting bolts in the following order. - Mounting bolt: Act : kgm - Install air intake connector.	**6.0 ~7.5 kgm** (Shop Manual Page. 50-32)				

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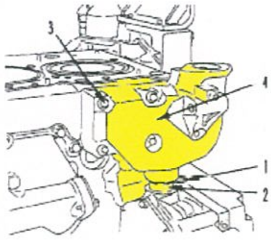
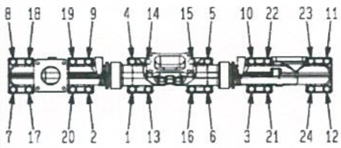
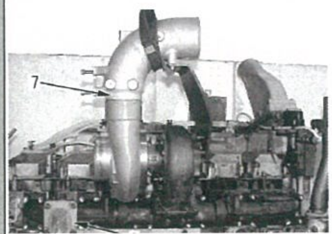
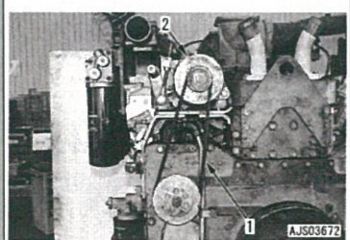
PT. SAPTAINDRA SEJATI

MAIN ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3				
	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC.DATE	28/08/2009
	ENGINE S/N *(new)*		REVISION NO.	06
	MACH.MODEL	HD 785-7	PAGE	11 / 12

No.	PARTS	CHECK	CONDITIONS TO BE CHECKED	SPEC.	SIGN	SKETCH DRAWING or
30	Water Pump		-Fit O-ring and install water pump assembly Water Pump: New / Reuse *** The mounting bolts of the water pump are used to tighten the gear case.** Tightening Mounting bolt: Act : kgm	**9,5 ~ 10,5 Kgm** (Shop Manual page. 00-32)		
31	Thermo-stat housing		-Fit O-ring to tube (21, then install to thermostat housing (4). -Fit gasket and install thermostat housing (4) with mounting bolts (3). -Move tube (2) down and install ring (l). Tightening Mounting bolt: Act : kgm	**6.7± 0.7 Kgm** (Shop Manual page. 00-32)		
32	Exhaust manifold & Turbo Charger	Exhaust manifold	- Fit the gasket and sling and install exhaust manifold and then tighten the mounting bolts in the following order - Mounting bolt: Act : kgm	**6.0 ~7.5 Kgm** (Shop Manual Page. 50-27)		
		Turbo Charger	- Fit the gasket and install turbocharger assembly - Turbo charger mounting bolt torque : Act : kgm **-<u>Turbocharger assy S/N:</u>** **RH:** **LH:** *-Fill oil to Turbocharger oil inlet for pre Lubricating when engine first running in Dyno(until fully on oil inlet hole)*	**4.5~5.0 Kgm** (Shop Manual Page. 13-51)		
		Alternator	-Install alternator(2) -Install auxiliary equipment belt(1) -Adjust tension of auxiliary equipment belt. *-adjust its deflection 20mmwhen the pushing force 6kgm,its applied to the center trought the finger.* Adjuster lock bolt: Act : kgm S/N Alternator:	**5.5 ~ 12.5 Kgm**		

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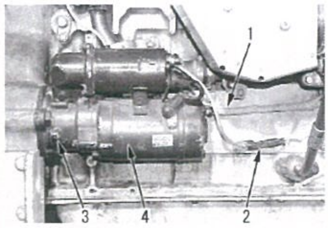
PT. SAPTAINDRA SEJATI

MAIN ASSEMBLY CHECK SHEET **ENGINE ASS'Y** KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC.DATE	28/08/2009
	ENGINE S/N *(new)*		REVISION NO.	06
	MACH.MODEL	HD 785-7	PAGE	12 / 12

		POINT			MEC	QA	SPV	PHOTOES
33	**Starting Motor**		- Fit the gasket and install starting motor assembly (4) with bolts (3) - Install wiring harness terminal B (1) and connector terminal R (2). mounting bolt torque: **-Starting motor assy S/N:** **Upper:** **Lower:** Act : kgm	**20 Kgm** (Reman Standard)				
Carry out a visual inspection of the engine prior to dyno testing								

Tightening Torque for Sensor	
Sensor name	**Tightening torque**
NE	4.9 – 7.9 Nm {0.5 – 0.8 kgm}
Engine speed	69 – 74 Nm{7 – 7.5 kgm} (Locknut)
Fuel temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
Boost pressure	23 ± 3 Nm {2.35 ± 0.3 kgm}
Intake air temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
Coolant temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
EGR gas pressure	23 ± 3 Nm {2.35 ± 0.3 kgm}
Engine oil pressure	23 ± 3 Nm {2.35 ± 0.3 kgm}
Atmospheric pressure	7.3 – 12.7 Nm {0.7 – 1.3 kgm} (Hexagon head bolt) 4.1 – 9.5 Nm {0.4 – 1.0 kgm} (Hexagon socket head bolt)
Fuel pressure	**Never disassemble.**

Notes :

ASSEMBLY BY	INSPECTED & APPROVED BY	
MECHANIC (MEC)	QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION

REV.02

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PLANT REBUILD CENTER

ENGINE ASS'Y

ENGINE PERFORMANCE TEST CHECKSHEET

KOMATSU ENGINE 12V140-3

MACHINE MODEL

HD 785-7

JOB NO:

ENGINE ASS'Y SECTION

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ENGINE PERFORMANCE TEST REPORT	DOCUMENT NO.	RC/QR05/EG/02/01
	PUBLICATION DATE	18/08/2009
	REVISION NO.	01
	PAGE	

CRITICAL ITEM CHECK

Name : QA INSPECTOR	Name : PROD. SUPERVISOR

WO NUMBER		START DATE / TIME	
ENGINE MODEL	SAA12V140E-3	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	HD 785-7	TEST BY	

RUNNING TEST

No	STANDARD (MANUAL)		ACTUAL TESTED	
	RPM	HP	RPM	HP
1	650	0		
2	1000	79		
3	1250	197		
4	1550	489		
5	1800	852		
6	1900	1200		

MEASUREMENT BLOWBY PRESSURE AT RATED SPEED (WITH MANUAL TEST)

STANDARD PRESSURE		ACTUAL		REMARK
MAXIMUM PRESSURE	300 mmH2O(2.94 Kpa)	1st		
SERVICE LIMIT VALUE	600 mmH2O(5.88 Kpa)	2nd		

	STANDARD	ACTUAL	REMARK
ALTERNATOR OUTPUT VOLTAGE	27.5 ~ 29.5		

MAINTENANCE STANDARD

SPEED (Rpm)		OIL Press. (Kg/cm²)	OIL Temp (°C)	WATER Press. (Kg/cm^2)	WATER Temp (°C)	BOOST PRESS (mmHg)	BLOW BY (mmH2O)	EXH TEMP (°C)
Low idle	650 ± 20	Min. 0.8	90 ~ 110	-	70~90	Min. 1250 at rated HP	Max 300 permisible 600	Max 650 at rated HP and Max. 700 at Max.torque
High idle	2250 ± 25	3.0 ~ 4.5						
POWER (HP) / RPM	1160 ~ 1240 / 1900 ± 50							
TORQUE (Kgm) / RPM	502~ 534 / 1350 ± 50							
EXHAUST TEMP DIFF	Maximum 10°C RH and LH Bank							

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ENGINE PERFORMANCE TEST REPORT	DOCUMENT NO.	RC/QR05/EG/02/01
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	REVISION NO.	01
	PAGE	

WO NUMBER		START DATE / TIME	
ENGINE MODEL	SAA12V140E-3	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	HD 785-7	TEST BY	

QUALITY PROBLEM

DESCRIPTION	REMARKS
WHAT (PROBLEM NAME)	
WHERE (PROBLEM LOCATION)	
WHEN (PROBLEM DATE)	
WHO (INSPECTED BY)	
HOW (PROBLEM EXPLAIN)	

CORRECTIVE ACTION :

PREVENTIVE ACTION :

REMARKS :

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ENGINE PERFORMANCE TEST REPORT	**DOCUMENT NO.**	RC/QR05/EG/02/01	
	PUBLICATION DATE	18/08/2009	
	REVISION NO.	01	
	PAGE		

CRITICAL ITEM CHECK

Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

WO NUMBER		**START DATE / TIME**	
ENGINE MODEL	SAA12V140E-3	**FINISH DATE / TIME**	
ENGINE SER.NO.		**DYNO.ARM LENGTH**	716 mm
MACHINE MODEL	HD 785-7	**TEST BY**	

RECHECK VALVE CLEARANCE AFTER LOAD

			PARAF	
			MEC	QA
Adjustment valve clearance	* Penentuan posisi top cylinder engine harus dicheck dan disetujui oleh QA. Step and method adjustment of value clearance following to the shop manual attached. Firing Order: R1-L1-R5-L5-R3-L3-R6-L6-R2-L2-R4-L4. Valve clearance: Intake valve : 0.35 ± 0.02 Exhaust valve : 0.57 ± 0.02 Tighten locknut to secure adjustment screw. Locknut: 5.4 ~ 6.6 kgm	Act : mm Act : mm Act : kgm		
Rocker Arm cover	Fit O ring and install rocker arm cover. Since the bolt at the front on the air intake manifold side is also used to secure the wiring harness clamp,tighten it temporary.			
	Mounting bolt : 3.0 ~ 3.5 kgm.	Act : kgm		

INSPECTION BY	**INSPECTION & APPROVAL BY**	
MECHANIC	**QUALITY ASSURANCE (QA)**	**GL/SPV PRODUCTION**

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

DELIVERY
INSPECTION SHEET
ENGINE

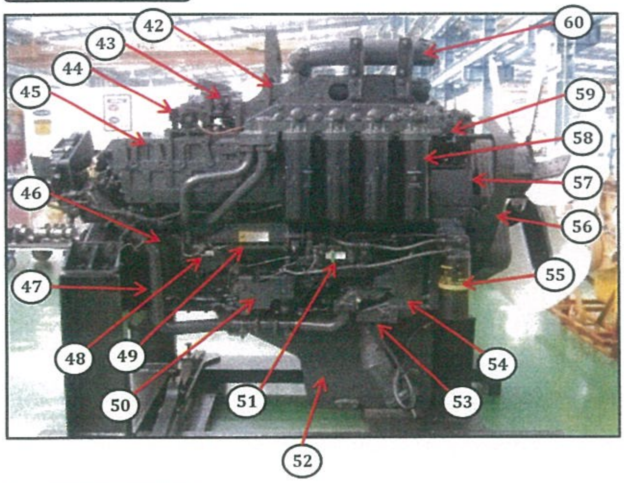
MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	2 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

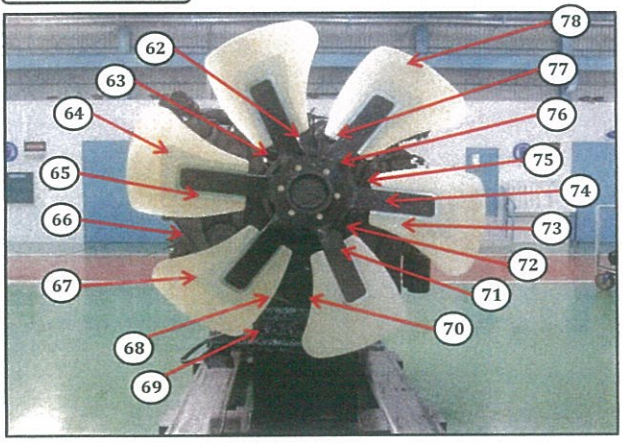
✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES

Right Side View



Front Side View



NO.	ITEM	INSPECTION CONDITION
42	Air Connector RH	
43	Exhaust Brake Assy RH	
44	Heater Switch Assy With Cover RH	
45	Air Intake Manifold RH	
46	Oil Cooler Assy	
47	Tube Cooling System Piping	
48	Fuel Pressure Sensor With Limiter Assy RH	
49	Common Rail Assy	
50	Priming Pump Assy With Cover RH	
51	Fuel Supply Pump Assy RH (P/N : 6219-71-1120)	
	S/N :	
52	Oil Pan	
53	Water Inlet Cover Plate	
54	Water Pump Assy	
55	Fuel Filter Assy	
56	Cover Safety Guard Alternator	
57	Tube, Breather With Hose RH	
58	Oil Filter Assy	
59	Alternator Assy (90A) (P/N : 600-825-9330)	
	S/N :	
60	Tube Air Intake Connection RH	
62	Plate Front Hanger	
63	Fan Drive With Pulley Assy	
64	Block Fuel Return With Sensor Assy RH & LH	
65	V-Belt Alternator	
66	Alternator Drive Pulley	
67	Tension Pulley Assy	
68	Vibration Damper Assy (P/N : 6215-31-8200)	
69	Front Support	
70	Crank Pulley Assy	
71	Pointer	
72	Yoke	
73	V-Belt Fan Set	
74	Blowbay Sensor Assy	
75	Water Temperature Sensor	
76	Thermostat Housing	
77	Tube Water Outlet (2 pcs)	
78	Cooling Fan	

Notes :

Inspected by,

Aproved by,

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

DELIVERY
INSPECTION SHEET
ENGINE

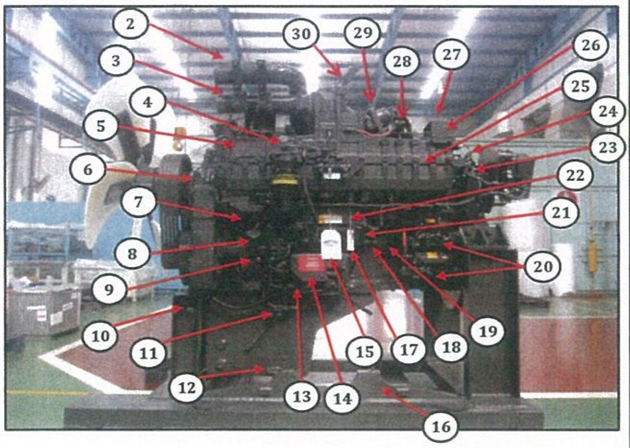
MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	1 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

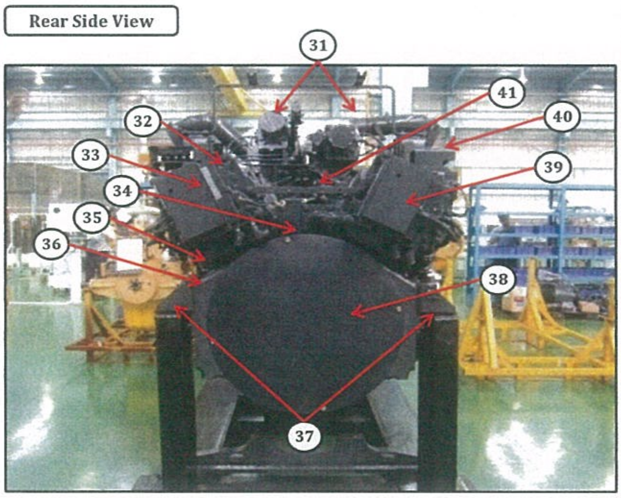
✓ : GOOD CONDITION   X : BAD CONDITION   ⊗ : NONE

SKETCH DRAWING OR PHOTOES

Left Side View



Rear Side View



NO.	ITEM	INSPECTION CONDITION
1	Painting (Y/B)	
2	Tube Air Intake Connection LH	
3	Air Connector LH	
4	Fuel Filter Assy LH	
5	Engine Name Plate	
6	Tube, Breather With Hose LH	
7	Fuel Supply Pump Assy LH (P/N : 6219-71-1110)	
	S/N :	
8	Tube Oil Filler Cap	
9	Gauge Oil (STD : H+ 0~10 mm)	
	Act : H +………………..mm	
10	Engine Stand (Original / Std)	
	Stand No :	
11	Oil Level Sensor Assy	
12	Drain Oil Plug	
13	Priming Pump Assy With Cover LH	
14	Warning Sticker Oil Level	
15	Bracket Mounting	
16	Quality passed Inspection Tag	
17	Caution Tag	
18	Oil Temperature Sensor	
19	Oil Pressure Sensor	
20	Starting Motor Assy (7.5 KW) (P/N : 600-813-7542)	
	Upper S/N :	
	Lower S/N :	
21	Fuel Pressure Sensor With Limiter Assy LH	
22	Common Rail Assy	
23	Boost Pressure Sensor	
24	Intake Temperature Sensor	
25	Air Intake Manifold LH	
26	Exhaust Temperature Sensor (4 pcs) With Bracket	
27	Customer Responsibilities Sticker	
28	Heater Switch Assy With Cover LH	
29	Exhaust Brake Assy LH	
30	Bracket Exhaust Brake Mounting	
31	Turbocharger Assy RH P/N : 6505-67-5030	
	S/N :	
	Turbocharger Assy LH P/N : 6505-67-5040	
	S/N :	
32	Heat Tab position	
33	Engine Controller Assy LH	
	S/N : ESN :	
	P/N Assy :	
34	Caution Flywheel Sticker	
35	Pick Up Revolution Sensor	
36	Flywheel Housing	
37	Rear Bracket Mounting (2 pcs)	
38	Flywheel (P/N : 6219-31-4110)	
39	Engine Controller Assy RH	
	S/N : ESN :	
	P/N Assy :	
40	Stiker Attention Installation Report	
41	Exhaust Manifold Assy RH & LH	

Notes :